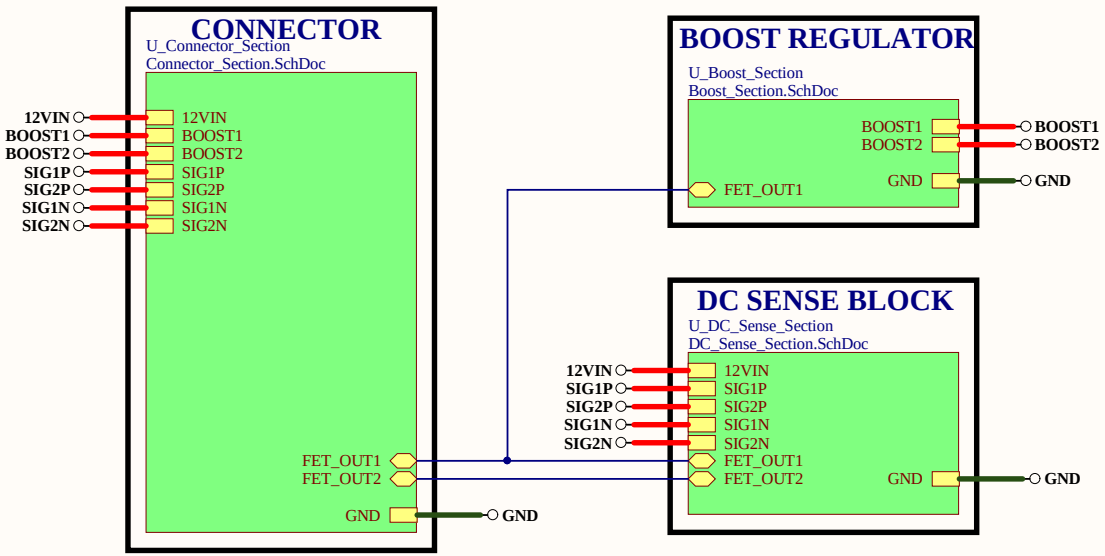
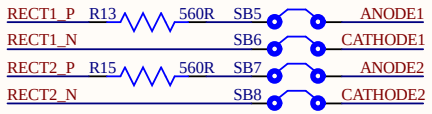

CONNECTOR
U_Connector_Section
Connector_Section.SchDoc
12VIN
BOOST1
BOOST2
SIG1P
SIG2P
SIG1N
SIG2N
12VIN
BOOST1
BOOST2
SIG1P
SIG2P
SIG1N
SIG2N
FET_OUT1
FET_OUT2
GND
GND
BOOST REGULATOR
U_Boost_Section
Boost_Section.SchDoc
BOOST1
BOOST2
GND
FET_OUT1
BOOST1
BOOST2
GND
DC SENSE BLOCK
U_DC_Sense_Section
DC_Sense_Section.SchDoc
12VIN
SIG1P
SIG2P
SIG1N
SIG2N
12VIN
SIG1P
SIG2P
SIG1N
SIG2N
FET_OUT1
FET_OUT2
GND
FET_OUT1
FET_OUT2
GND
GND

CONNECTING SENSE OUTPUTS

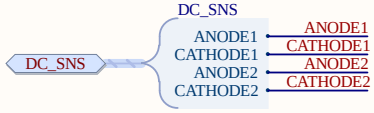


Connect these solder bridges to use the rectified signal for USB input sensing.

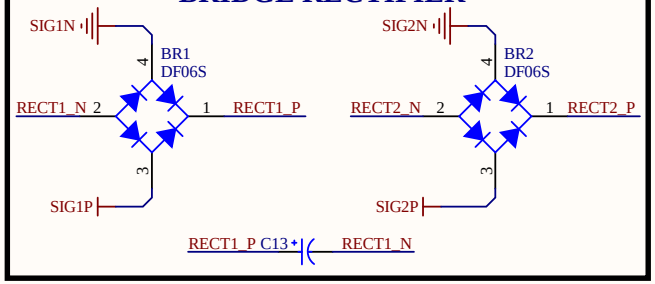
LED POWER IN SIGNAL



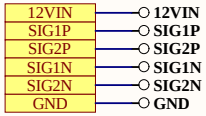
DC SENSE OUTPUT



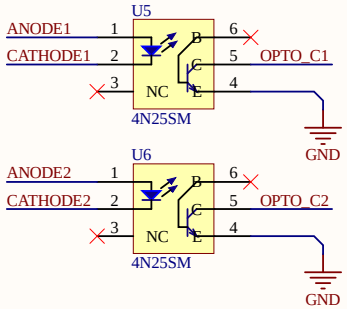
BRIDGE RECTIFIER



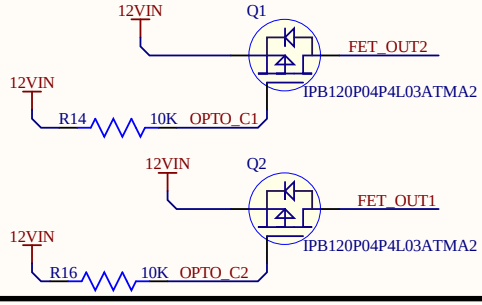
POWER SIGNALS



SENSE DETECT OPTO-ISOLATOR

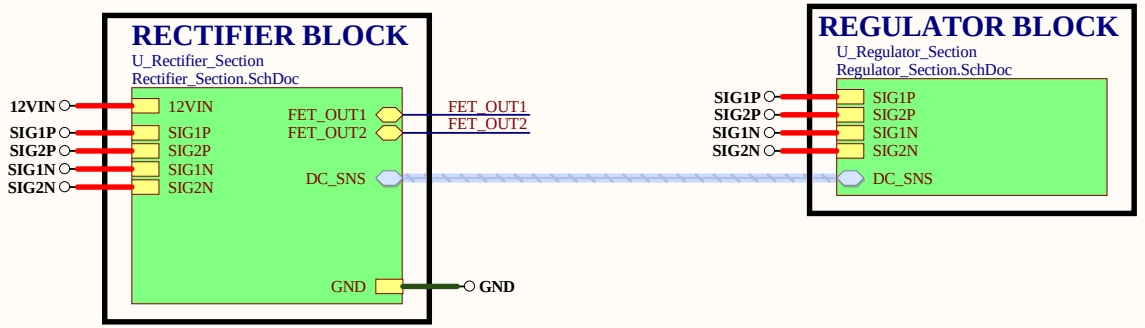


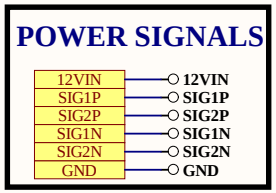
MOSFET FOR POWER STRIP CONTROL



USB Sense signal SIG1 and SIG2 will be controlling Q2 and Q1, respectively.
FETOUT will be used to power up LED strips.

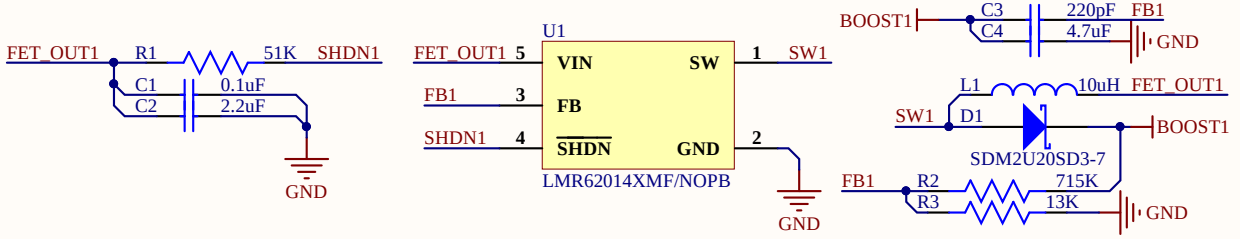

FET_OUT1
FET_OUT2
FET_OUT1
FET_OUT2


RECTIFIER BLOCK
U_Rectifier_Section
Rectifier_Section.SchDoc
12VIN
SIG1P
SIG2P
SIG1N
SIG2N
12VIN
SIG1P
SIG2P
SIG1N
SIG2N
FET_OUT1
FET_OUT2
DC_SNS
GND
FET_OUT1
FET_OUT2
GND
REGULATOR BLOCK
U_Regulator_Section
Regulator_Section.SchDoc
SIG1P
SIG2P
SIG1N
SIG2N
SIG1P
SIG2P
SIG1N
SIG2N
DC_SNS

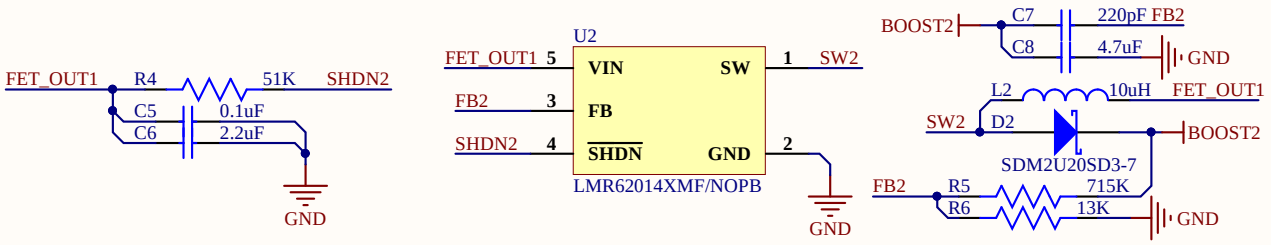

POWER SIGNALS
12VIN
SIG1P
SIG2P
SIG1N
SIG2N
GND
12VIN
SIG1P
SIG2P
SIG1N
SIG2N
GND

FET_OUT1 FET_OUT1

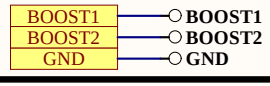
12V BOOST REGULATOR FOR RGB STRIP01



12V BOOST REGULATOR FOR RGB STRIP02



POWER SIGNALS



Design Notes for Boost Regulator Circuits

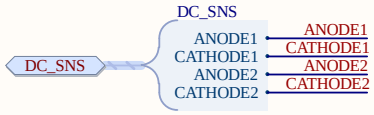
The resistor calculation is done from the formula reference pg.10 of the datasheet
R1 = R2 X (VOUT/1.23 − 1)

The output from the regulator is 12.65V to allow diode drop compensation and thus acheieve stable 12V for the rest of circuit. The diode drop is 525mV.

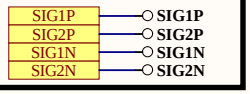
From the formula:
VOUT = (R1 / R2) * (1.23 - 1) = (715K / 13K) * (1.23 - 1)
VOUT = 12.65V - 0.525V (Diode Drop)
VOUT = 12.125V

Title		
Size: A	Number	Revision
Date: 11/13/2021		Sheet of
File: C:\Users\..\Boost_Section.SchDoc		Drawn By:

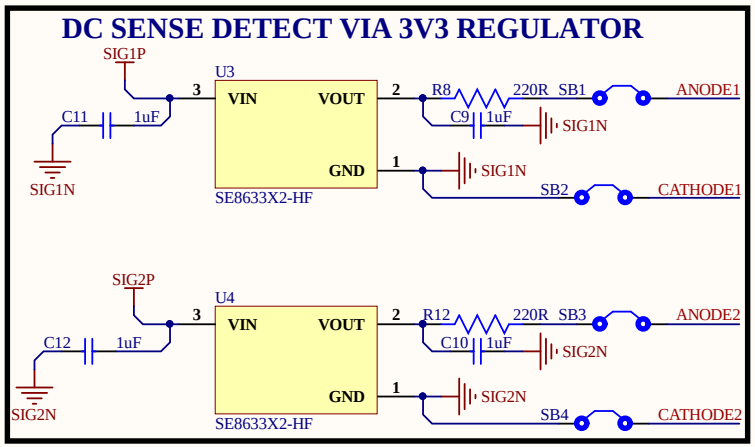
DC SENSE OUTPUT



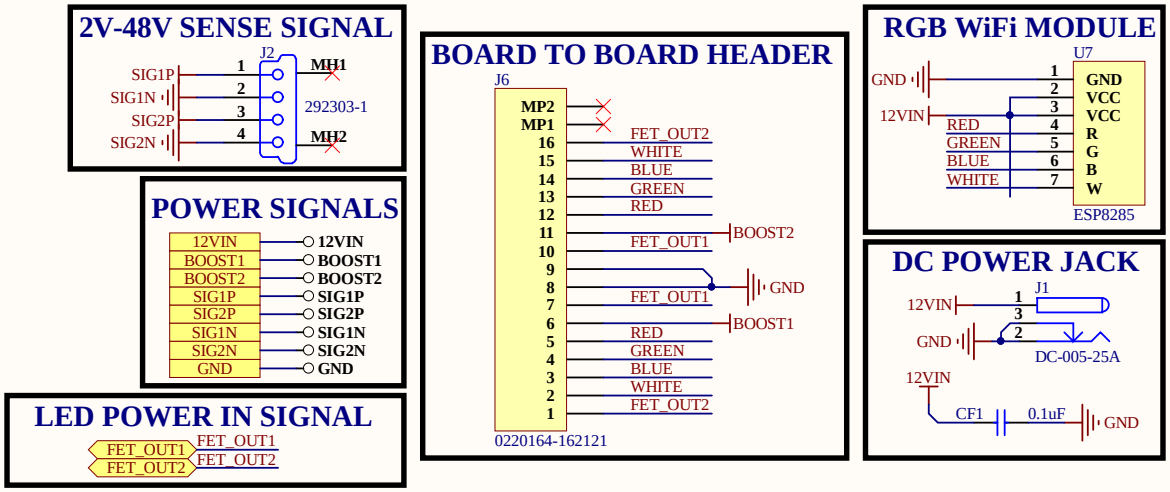
POWER SIGNALS



DC SENSE DETECT VIA 3V3 REGULATOR



Connect these solder bridges to use the regulators for USB input sensing.


2V-48V SENSE SIGNAL
J2
SIG1P
SIG1N
SIG2P
SIG2N
1
2
3
4
MH1
MH2
292303-1
POWER SIGNALS
12VIN
BOOST1
BOOST2
SIG1P
SIG2P
SIG1N
SIG2N
GND
LED POWER IN SIGNAL
FET_OUT1
FET_OUT2
BOARD TO BOARD HEADER
J6
MP2
MP1
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
FET_OUT2
WHITE
BLUE
GREEN
RED
BOOST2
FET_OUT1
GND
BOOST1
0220164-162121
RGB WiFi MODULE
U7
GND
12VIN
RED
GREEN
BLUE
WHITE
VCC
R
G
B
W
ESP8285
DC POWER JACK
J1
DC-005-25A
CF1
0.1uF